1. Growth of Data Processing Operations in an Algorithm

An algorithm performs a certain number of operations to process input data. As the input size increases, the number of operations performed by the algorithm also increases. The growth of these operations can be represented using a recurrence relation.

Let T(n) represent the number of operations performed by an algorithm for an input size n.

Assume that the algorithm processes one data item at a time. For an input of size n, it performs one operation and then processes the remaining n − 1 data items.

Step 1: Define the operation

Let:

$$T(n) = \text{Number of operations for input size } n$$

For every data item, the algorithm performs **1 operation**.

Step 2: Form the recurrence relation

For input size n, the algorithm performs:

- $T(n-1)$operations for the remaining $n-1$items.
- 1 additional operation for the current item.

Therefore,

$$\boxed{T(n) = T(n-1) + 1}$$

Step 3: Give the initial condition

For one data item, only one operation is required.

$$\boxed{T(1) = 1}$$

Step 4: Calculate for different input sizes

For $n = 2$:

$$T(2) = T(1) + 1$$
$$= 1 + 1 = 2$$

For $n = 3$:

$$T(3) = T(2) + 1$$
$$= 2 + 1 = 3$$

For $n = 4$:

$$T(4) = T(3) + 1$$
$$= 3 + 1 = 4$$

For $n = 5$:

$$T(5) = T(4) + 1$$
$$= 4 + 1 = 5$$

Step 5: Observe the growth

Therefore, the number of operations is:

$$1,2,3,4,5, \dots, n$$

Hence,

$$\boxed{T(n) = n}$$

The number of operations grows **linearly** with the input size.

Therefore, the time complexity of the algorithm is:

$$\boxed{O(n)}$$

Final Answer

The recurrence relation representing the growth of data processing operations is:

$$\boxed{T(n) = T(n-1) + 1,\ T(1) = 1}$$

Its solution is:

$$\boxed{T(n) = n}$$

Thus, the algorithm has linear growth and its time complexity is O(n).

2. Recurrence Relation and Data Transfer Analysis of Tower Of Hanoi

Introduction

The Tower of Hanoi is a classical problem used in algorithm analysis. It consists of three rods and a set of disks of different sizes. The disks are initially arranged on one rod in decreasing order of size, with the largest disk at the bottom. The objective is to transfer all the disks from the source rod to the destination rod using the auxiliary rod.

Rules of the Tower of Hanoi

- Only one disk can be moved at a time.
- Only the top disk from a rod can be moved.
- A larger disk must never be placed on a smaller disk.
- All disks must finally be transferred from the source rod to the destination rod.

Tower of Hanoi as a Data Transfer Model

The three rods can be considered as storage locations. The source rod represents the original storage location, the destination rod represents the new storage location, and the auxiliary rod provides temporary storage. Each disk can represent a unit or block of data with a restriction on the order in which it can be transferred.

For n disks, the problem can be divided into three steps: first transfer the top $n-1$ disks to the auxiliary rod, then transfer the largest disk to the destination rod, and finally transfer the $n-1$ disks from the auxiliary rod to the destination rod.

Recurrence Relation

Let $T(n)$ be the minimum number of operations required to transfer n disks.

To transfer n disks:

1. Transfer $n-1$ disks from source to auxiliary: $T(n-1)$ operations.

2. Transfer the largest disk from source to destination: 1 operation.

3. Transfer $n-1$ disks from auxiliary to destination: $T(n-1)$ operations.

Therefore, the recurrence relation is:

$$T(n) = 2T(n-1) + 1$$

Base condition:

$$T(1) = 1$$

TOWER OF HANOI – SOLUTION

Solving the Recurrence

Starting with $T(n) = 2T(n-1) + 1$:

$T(n) = 2[2T(n-2) + 1] + 1$

$= 2^2T(n-2) + 2 + 1$

Continuing the expansion:

$T(n) = 2^3T(n-3) + 2^2 + 2 + 1$

After $n-1$ expansions:

$T(n) = 2^{(n-1)}T(1) + (2^{(n-1)} - 1)$

Since $T(1) = 1$:

$T(n) = 2^{(n-1)} + 2^{(n-1)} - 1$

Therefore, the closed-form solution is:

$T(n) = 2^n - 1$

Number of Operations for Different Disks

Number of Disks (n)	Minimum Operations T(n) = 2^n − 1
1	1
2	3
3	7
4	15
5	31
6	63
7	127

Conclusion

The Tower of Hanoi provides a simple model for studying data transfer and recursive algorithms. The number of transfer operations follows the recurrence relation $T(n) = 2T(n-1) + 1$, whose solution is $T(n) = 2^n - 1$. Therefore, even a small increase in the number of disks causes a large increase in the required operations. This example clearly demonstrates the importance of recurrence relations in algorithm analysis.

3.Investigate A Digital System Used In Computer System

Digital System Selected – Computer Login System

A Computer Login System is a digital system used to control access to a computer. It checks whether the entered login details are correct.

Consider three inputs:

$$A = \text{Username is correct}$$

$$B = \text{Password is correct}$$

$$C = \text{Security verification is correct}$$

The output is:

$$Y = \text{Access Granted}$$

The system gives access only when all three conditions are correct.

Represent The Functionality Using Boolean Algebra

The login system uses an AND operation because all three conditions must be satisfied.

Therefore, the Boolean function is:

$$Y = A \cdot B \cdot C$$

Here,

· represents the AND operation.

A, B and C are the input conditions.

Y is the output.

If all three inputs are 1:

$$A = 1, B = 1, C = 1$$

$$Y = 1 \cdot 1 \cdot 1 = 1$$

Therefore, Access Granted.

If any one of the inputs is 0, the output becomes 0, which means Access Denied.

TRUTH TABLE

A B C $Y = A \cdot B \cdot C$

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

From the truth table, the output is 1 only when A = 1, B = 1 and C = 1.

Therefore, $Y = \Sigma m(7)$

Minimize The Boolean Function Using K-Map

For three variables A, B and C, the K-Map contains 8 cells.The output is 1 only at minterm 7.

K-Map:

AB\C	0	1
00	0	0
01	0	0
11	0	1
10	0	0

There is only one 1 in the K-Map, so it cannot be combined with another 1.

Therefore, the minimized Boolean function is:

$$Y = A \cdot B \cdot C$$

Final Result

The Computer Login System was represented using Boolean Algebra as:

$$Y = A \cdot B \cdot C$$

After applying the K-Map, the minimized Boolean function is also

$$Y = A \cdot B \cdot C$$

Thus, the computer grants access only when the username, password and security verification are all correct.

The process can be understood as:

Digital System → Boolean Function → Truth Table → K-Map → Minimized Function

Final Answer:

$$Y = A \cdot B \cdot C$$